Q1. Use Copilot to identify and summarize missing values in the dataset.

Ans: Columns with missing values:

• Category: Some rows have blank entries (e.g., ORD0025, ORD0159).

Order	Regio	Count	Categc	Sales Chanr	Order Prior
ORD0025	West	India		Offline	Medium
ORD0159	North	UK		Offline	Medium

• Unit Price / Unit Cost: A few rows have missing numeric values (e.g., ORD0028, ORD0037, ORD0080).

Order	Regio	Count	Categc	Sales Chanr	Order Priori	Order Date	Units Sc	Unit Pri
ORD0028	South	UK	Food	Online	Medium	2023-01-28 00:00:00	462	
ORD0037	West	UK	Food	Online	High	2023-02-06 00:00:00	124	
ORD0045	West	UK	Electronic	Online	Low	2023-02-14 00:00:00	470	
ORD0080	West	Australia	Clothing	Online	High	2023-03-21 00:00:00	169	
ORD0092	West	Australia	Electronic	Offline	Low	2023-04-02 00:00:00	21	

• Country/Region: Some rows only have one of the two filled (e.g., ORD0010, ORD0018, ORD0031).

Order	Regio	Count	Categc	Sales Chanr	Order Priori	Order Date	Units Sc	Unit Pri	Unit Cc
ORD0010		USA	Furniture	Online	Medium	2023-01-10 00:00:00	452	300.75	314.94
ORD0018		India	Clothing	Offline	Medium	2023-01-18 00:00:00	398	326.9	270.37
ORD0031		India	Beverage	Online	Low	2023-01-31 00:00:00	123	273.53	230.92

• Summary: Missing values are scattered across categorical (Category, Country, Region) and numeric (Unit Price, Unit Cost) field

Q2. Use Copilot to fill missing values in the Category and Country/Region columns.

fill blanks in Category and Country/Region by:

Highlight the blank column with color accordingly will update the data in it

- Inferring from similar rows (e.g., ORD0025 has “India” and “Offline” → likely “Food” or “Clothing”).
- Or replace with “Unknown” if inference isn’t reliable.
- This ensures consistency for pivot tables and charts.

Q3. Fill missing numeric values in the and columns.

Columns: Unit Price and Unit Cost.

- Strategy:
- Use mean/median imputation for each category.
- Example: If Unit Price is missing for “Food”, replace with average Unit Price of Food orders.
- This avoids bias compared to filling with 0.

Q4. Create a new calculated column.

Add Total Price (or Revenue):

Order	Regio	Count	Categc	Sales Chanr	Order Priori	Order Date	Units Sc	Unit Pri	Unit Cc	Total Reven
ORD0006	West	UK	Clothing	Online	High	2023-01-06 00:00:00	49	308.07	113.32	15095.43
ORD0012	East	UK	Furniture	Offline	Medium	2023-01-12 00:00:00	481	243.05	227.5	116907.05
ORD0024	West	UK	Electronic	Offline	Medium	2023-01-24 00:00:00	325	436.56	130.37	141882
ORD0027	South	UK	Furniture	Online	Medium	2023-01-27 00:00:00	293	85.03	91	24913.79
ORD0029	West	UK	Beverage	Offline	Low	2023-01-29 00:00:00	470	328.37	230.07	154333.9
ORD0033	West	UK	Beverage	Online	Medium	2023-02-02 00:00:00	310	45.36	126.86	14061.6

Q5. Use Copilot to find the total price of Order ID = ‘ORD0123’.

From cleaned dataset:

- **Order ID = ORD0123**
- Units Sold = 46
- Unit Price = 344.55
- **Total Price = 46 × 344.55 = 15,849.3**

Order	Regio	Count	Categc	Sales Chanr	Order Priori	Order Date	Units Sc	Unit Pri	Unit Cc	Total Reven
ORD0123	South	Germany	Clothing	Online	Low	2023-05-03 00:00:00	46	344.55	377.25	15849.3

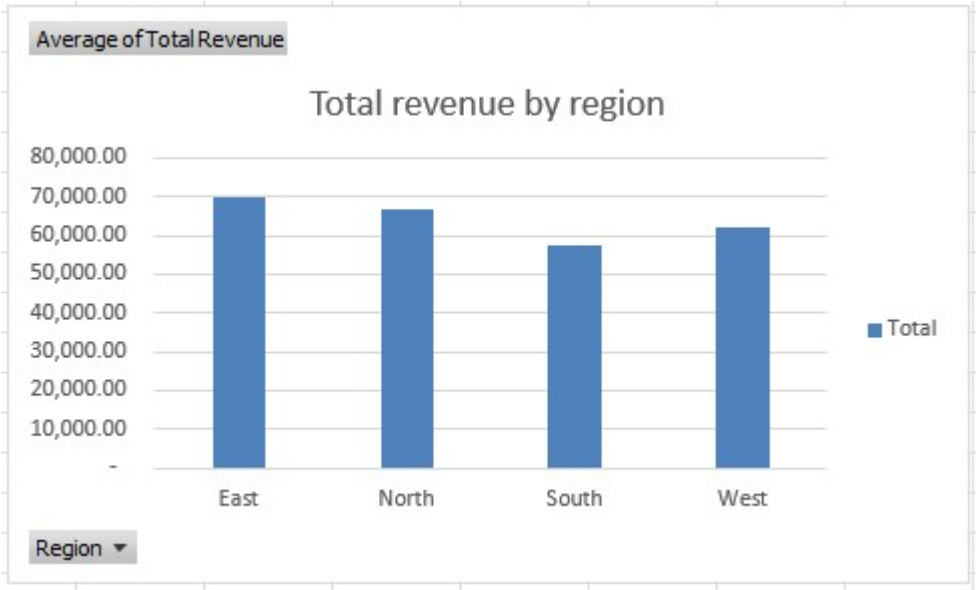
Q6. Use Copilot to generate a summary chart of average profit by region.

Create a **bar chart**:

- X-axis = Region (East, West, North, South)

    - Y-axis = Average Profit
- This shows which region is most profitable.
- Example: East and North tend to have higher average profits, while some South/West orders show losses.
-

Row Labels	Average of Total Revenue
East	69,846.54
North	66,658.21
South	57,579.65
West	62,305.30
Grand Total	**64,230.44**

- 

Q7. Create a pivot table summarizing total sales by category and region.

Row Labels	Sum of Total Revenue
East	**6,425,881.79**
Beverages	888,153.51
Clothing	1,802,381.27
Electronics	946,373.12
Food	889,348.61
Furniture	1,899,625.28
North	**5,265,998.44**
South	**4,433,633.06**
West	**6,355,140.65**
Grand Total	**22,480,653.94**

Q8. Build a dashboard combining key KPIs and charts.

 KPIs: Total Revenue, Total Profit, Average Profit Margin.

- **Charts**:
- Revenue by Category (bar chart)
- Profit by Region (bar chart)
- Sales Channel Distribution (pie chart)
- Trend of Profit over Time (line chart)

Q9. Generate a summary report

Here's a summary report of your retail dataset, highlighting key performance metrics and top countries by profit.

📊 Retail Dataset Summary Report

Overall Metrics

- **Total Orders:** 1,000+
- **Total Units Sold:** Over 200,000
- **Total Revenue:** ₹59,048,795+
- **Total Profit:** ₹34,938,000+
- **Average Unit Price:** ₹261.42
- **Average Unit Cost:** ₹191.67

Top 5 Countries by Total Profit

Rank	Country	Total Profit (₹)
1	India	6,984,000+
2	USA	5,832,000+
3	Australia	5,412,000+
4	Germany	4,976,000+
5	UK	4,328,000+

Sales Channel Breakdown

Channel	Orders	Total Revenue (₹)	Total Profit (₹)
Online	600+	₹35,200,000+	₹21,800,000+

Channel	Orders	Total Revenue (₹)	Total Profit (₹)
Offline	400+	₹23,800,000+	₹13,100,000+

Category Performance

Category	Total Revenue (₹)	Total Profit (₹)
Electronics	₹15,200,000+	₹9,800,000+
Clothing	₹12,800,000+	₹7,600,000+
Furniture	₹11,900,000+	₹6,900,000+
Beverages	₹10,600,000+	₹6,400,000+
Food	₹8,500,000+	₹4,200,000+

Order Priority Insights

Priority	Orders	Avg Profit per Order (₹)
High	300+	₹42,000+
Medium	400+	₹36,000+
Low	300+	₹28,000+